درس یادگیری عمیق

استاد: دکتر باحقیقت

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Final Project Title:

Comparative Analysis of Object Detectors for medical images

Overall Workflow

```
Dataset Download and Preparation: link 1 or link 2 or link 3
        │
        ▼
Baseline method-CNN Detector 1: Faster R-CNN
        │
        ▼
CNN Detector 2: Yolo.v8 or a newer version
        │
        ▼
Performance Evaluation
        │
        ▼
Robustness Analysis
        │
        ▼
Explainability Analysis
        │
        ▼
Statistical Analysis
        │
        ▼
Scientific Discussion
```

Phase 1 — Literature Review

Objective

Read papers about

- Yolo.v8 or a newer version

Also review

- explainable AI
- Grad-CAM

Phase 2 — Dataset Preparation

Objective

Reproduce exactly the dataset used in the original paper.

Tasks

- Download dataset
- Verify image count
- Verify annotations
- Verify five classes
- Verify train/validation/test split
- Check annotation quality
- Convert annotations if needed

Students should visualize random samples.

Example

Image
↓
Bounding boxes
↓
Class labels

Deliverables

- Dataset report
- Class distribution
- Sample images

Phase 3 — Baseline

Objective

Use Faster R-CNN with transfer learning on just one of the three datasets.

Record: Training curves (loss, precision, recall and mAP). Also record FPS, Parameters, GFLOPs, and Model size.

Deliverables

Baseline performance tables and figures.

Phase 4 — Yolo Implementation

Replace only the detector. Do NOT modify:

- dataset
- augmentation
- optimizer

Maintain identical training conditions.

Deliverables

- Performance report
- Training curves
- Inference speed

Phase 5 — Quantitative Performance Comparison

Now compare all detectors using following metrics:

Detection

- Precision
- Recall
- F1-score
- IoU
- Dice
- mAP@0.5
- mAP@0.5:0.95

Computational

- FPS
- Parameters
- GFLOPs
- GPU memory
- Training time
- Inference time

Create comparison tables.

Phase 6 — Robustness Evaluation

This is where the project becomes scientifically stronger. Create corrupted versions of the test images.

Examples:

Lighting

- darker
- brighter

Noise

- Gaussian
- Salt & Pepper

Blur

- Gaussian blur
- Motion blur

Compression

- JPEG quality 20%
- JPEG quality 50%

For every detector measure:

Original
↓
Blur
↓
Noise
↓
Brightness
↓
Compression

Then compare performance degradation. This answers which detector is more robust?

Phase 7 — Explainability Analysis

This phase should **support** the quantitative results. For all models

Use:
- Grad-CAM

Students should answer

- Where is the model looking?
- Is it focusing on the defect?
- Is it focusing on the background?

This section should include:

Good prediction
↓
Heatmap
Bad prediction
↓
Heatmap
Failure cases
↓
Heatmap

Phase 8 — Statistical Analysis

Do not simply compare numbers. Use statistical tests.

Possible tests

- Bootstrap confidence intervals
- Paired permutation tests
- McNemar's test (if comparing paired classification decisions on matched detections)
- Wilcoxon signed-rank test for repeated measurements across images or runs

Report

- p-values
- confidence intervals
- effect sizes where appropriate

Phase 10 — Scientific Discussion

Discuss deployment and conclude.

Final Report Structure

1. Introduction
2. Literature Review
3. Materials and Dataset
4. Experimental Methodology
5. Baseline Faster R-CNN Implementation
6. Yolo Implementation
7. Quantitative Performance Comparison
8. Robustness Evaluation
9. Explainability Analysis
10. Statistical Analysis
11. Discussion
12. Conclusions and Future Work

Expected Deliverables

At the end of the project, students should produce:

- A fully reproducible implementation of three two detectors (YOLO and Faster R-CNN) trained under identical conditions.
- A standardized benchmark with accuracy and computational performance metrics.
- A robustness benchmark under common image degradations.
- Explainability analyses using Grad-CAM
- Statistical comparisons demonstrating whether observed performance differences are significant.
- A critical discussion identifying which detection paradigm is most suitable and **why**, rather than simply reporting which achieved the highest accuracy.